

Combining Covariate Adjustment with Group Sequential and Information-Adaptive Designs to Improve Randomized Trial Efficiency

RISW, September 2026

Plan

- 1 Background, Problem Setting and Set Up
- 2 Proposal: Combining Covariate Adjustment and GSDs
- 3 Information-Adaptive Design
- 4 Simulation Study
- 5 Discussion

Covariate Adjustment

- **Covariate adjustment** is a statistical analysis method with high potential to **improve precision** for many trials.
 - **Pre-planned** adjustment for baseline variables when estimating **average treatment effect**.
 - Estimand is same as when using unadjusted estimator (e.g., difference in means).
 - **Goal**: avoid making any model assumptions beyond what's assumed for unadjusted estimator (**robustness to model misspecification**).

(e.g., Koch et al., 1998; Yang and Tsiatis, 2001; Rubin and van der Laan, 2008; Tsiatis et al., 2008; Moore and van der Laan, 2009b,a; Zhang, 2015; Jiang et al., 2018; Benkeser et al., 2020)

Group Sequential Designs

- A commonly used type of clinical trial design that involves **pre-planned interim analyses**
 - where the trial can be **stopped early** for efficacy or futility.
- Prevalent in confirmatory clinical trials for **ethical and efficiency reasons** as they potentially **save time and resources** by allowing early termination of the trial.

Problem Setting

- **Combination** of covariate adjustment and group sequential designs has the **potential to offer the benefits of both methods**:
 - using covariate adjusted estimators at interim and final analyses of a group sequential design.
- Several **challenges** involved in combining these two approaches.

Potential Challenges

- 1 Many covariate adjustment methods are **incompatible with 'standard' group sequential designs** (GSDs).

Potential Challenges

- 1 Many covariate adjustment methods are **incompatible with 'standard' group sequential designs** (GSDs).
 - This is the case when models used to construct the estimators are misspecified.

Potential Challenges

- 1 Many covariate adjustment methods are **incompatible with 'standard' group sequential designs (GSDs)**.
 - This is the case when models used to construct the estimators are misspecified.
- 2 The **uncertainty** at the design stage about the **amount of precision gain and corresponding sample size reduction**.

Potential Challenges

- 1 Many covariate adjustment methods are **incompatible with 'standard' group sequential designs** (GSDs).
 - This is the case when models used to construct the estimators are misspecified.
- 2 The **uncertainty** at the design stage about the **amount of precision gain and corresponding sample size reduction**.
 - Solution: information-adaptive designs.

Endpoints, Estimands and Estimators

- The proposal works for **all types of common outcomes**
 - e.g., continuous, binary, ordinal, and time-to-event
- The proposal accommodates **any estimand**, including
 - risk difference, relative risk and odds ratio for binary outcomes,
 - difference in restricted mean survival times and relative risk for time-to-event outcomes,
 - ...
- The proposal will be applicable to **any (adjusted and unadjusted) estimator** as long as it is regular and asymptotically linear (RAL) and consistent for the estimand of interest.
 - e.g., *G-computation* estimator (as suggested in the recent FDA draft guidance on covariate adjustment)

Endpoints, Estimands and Estimators: Example

- Primary endpoint: binary.

Endpoints, Estimands and Estimators: Example

- Primary endpoint: binary.
- Estimand: $\theta = E(Y|A = 1) - E(Y|A = 0)$.

Endpoints, Estimands and Estimators: Example

- Primary endpoint: binary.
- Estimand: $\theta = E(Y|A = 1) - E(Y|A = 0)$.
- Estimator: G-computation/Standardization

1 Fit logistic regression model for

$$P(Y = 1|A, X) = \text{logit}^{-1}(\gamma_0 + \gamma_1 A + \gamma_2 X).$$

2 Compute standardized estimators for treatment specific means

- $\hat{E}(Y|A = 1) = \frac{1}{n} \sum_{i=1}^n \text{logit}^{-1}(\hat{\gamma}_0 + \hat{\gamma}_1 + \hat{\gamma}_2 X_i)$

- $\hat{E}(Y|A = 0) = \frac{1}{n} \sum_{i=1}^n \text{logit}^{-1}(\hat{\gamma}_0 + \hat{\gamma}_2 X_i)$

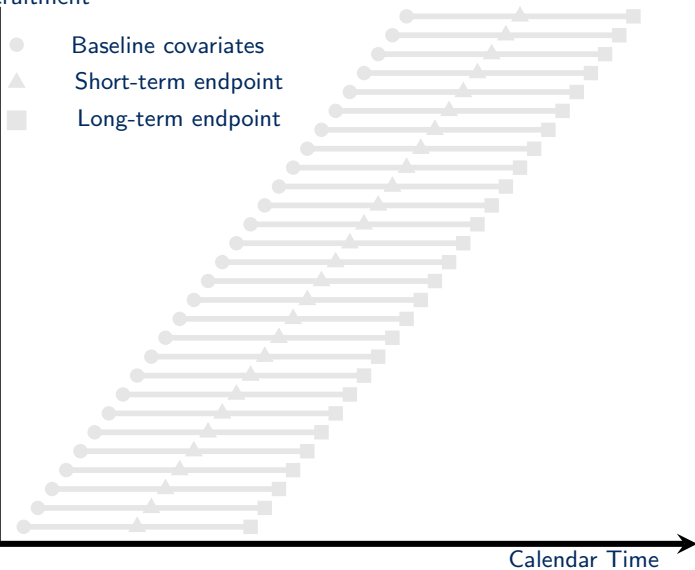
3 Calculate $\hat{\theta} = \hat{E}(Y|A = 1) - \hat{E}(Y|A = 0)$

Plan

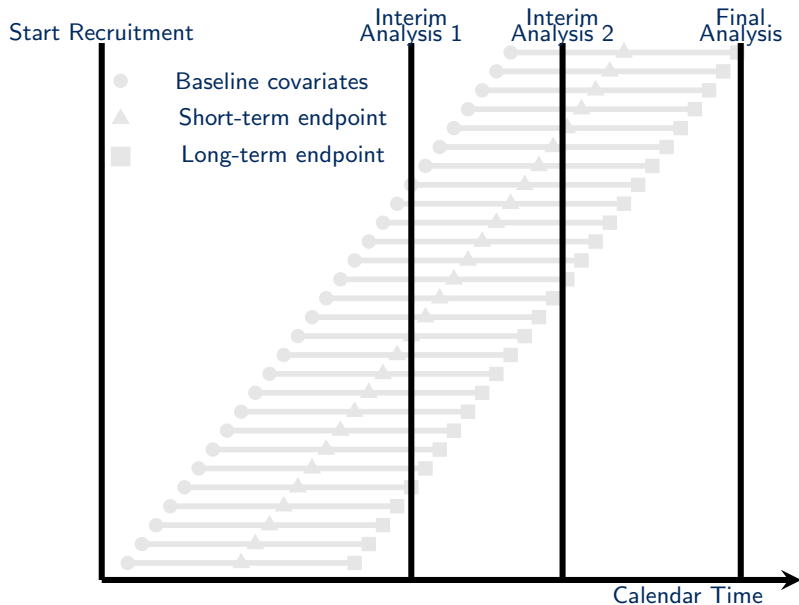
- 1 Background, Problem Setting and Set Up
- 2 Proposal: Combining Covariate Adjustment and GSDs**
- 3 Information-Adaptive Design
- 4 Simulation Study
- 5 Discussion

Group Sequential Design

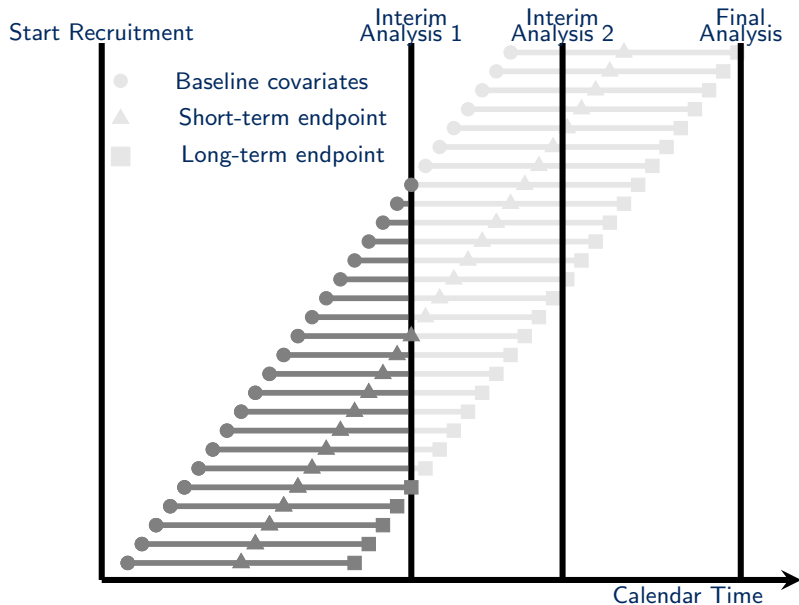
Start Recruitment



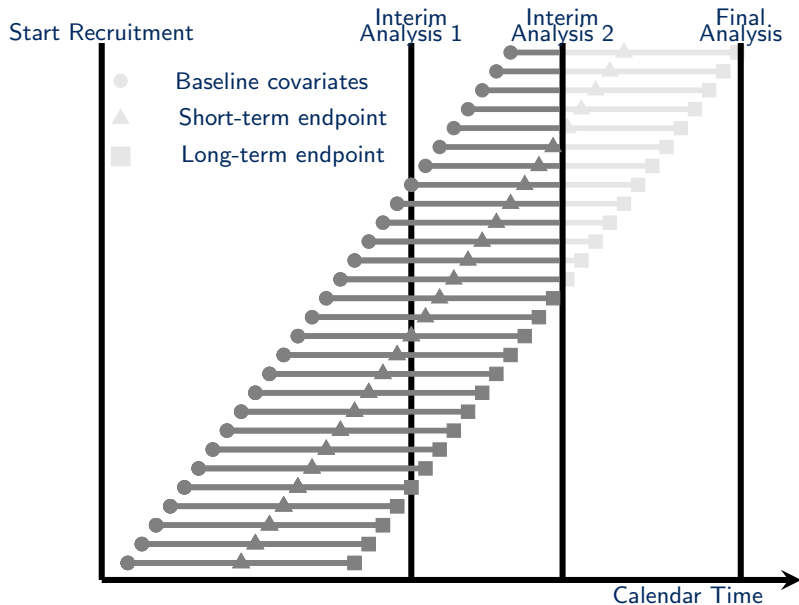
Group Sequential Design



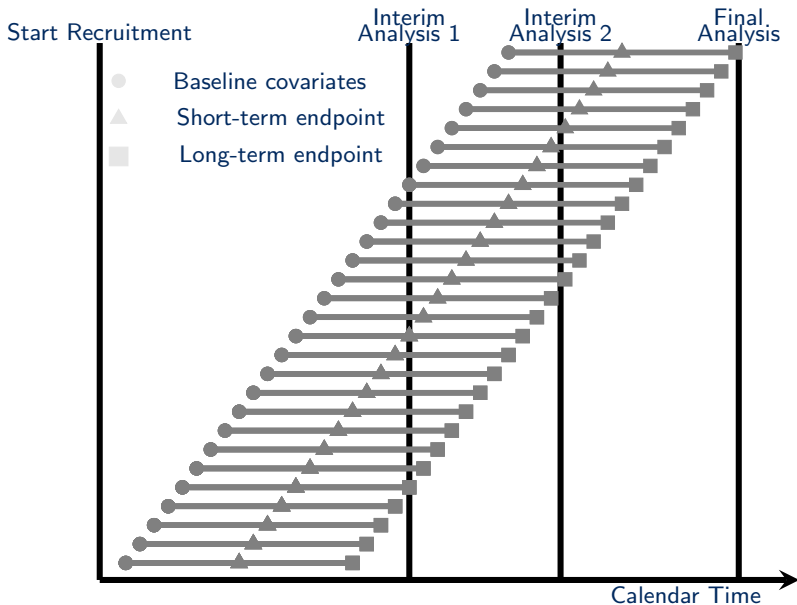
Group Sequential Design



Group Sequential Design



Group Sequential Design



Standardization Estimator at Interim Analyses

- Primary endpoint: binary.
- Estimand: $\theta = E(Y|A = 1) - E(Y|A = 0)$.
- Estimator: Standardization/G-computation

Standardization Estimator at Interim Analyses

- Primary endpoint: binary.
- Estimand: $\theta = E(Y|A = 1) - E(Y|A = 0)$.
- Estimator: Standardization/G-computation

At time t_k :

Standardization Estimator at Interim Analyses

- Primary endpoint: binary.
- Estimand: $\theta = E(Y|A = 1) - E(Y|A = 0)$.
- Estimator: Standardization/G-computation

At time t_k :

- 1 Fit logistic regression model for

$$P(Y = 1|A, X) = \text{logit}^{-1}(\gamma_0 + \gamma_1 A + \gamma_2 X),$$

in participants with complete follow up.

Standardization Estimator at Interim Analyses

- Primary endpoint: binary.
- Estimand: $\theta = E(Y|A = 1) - E(Y|A = 0)$.
- Estimator: Standardization/G-computation

At time t_k :

- 1 Fit logistic regression model for

$$P(Y = 1|A, X) = \text{logit}^{-1}(\gamma_0 + \gamma_1 A + \gamma_2 X),$$

in participants with complete follow up.

- 2 Compute standardized estimators for treatment specific means

- $\hat{E}_{t_k}(Y|A = 1) = \frac{1}{n'} \sum_{i=1}^{n'} \text{logit}^{-1}(\hat{\gamma}_0 + \hat{\gamma}_1 + \hat{\gamma}_2 X_i)$

- $\hat{E}_{t_k}(Y|A = 0) = \frac{1}{n'} \sum_{i=1}^{n'} \text{logit}^{-1}(\hat{\gamma}_0 + \hat{\gamma}_2 X_i)$

in all n' recruited patients.

Standardization Estimator at Interim Analyses

- Primary endpoint: binary.
- Estimand: $\theta = E(Y|A = 1) - E(Y|A = 0)$.
- Estimator: Standardization/G-computation

At time t_k :

- 1 Fit logistic regression model for

$$P(Y = 1|A, X) = \text{logit}^{-1}(\gamma_0 + \gamma_1 A + \gamma_2 X),$$

in participants with complete follow up.

- 2 Compute standardized estimators for treatment specific means

- $\hat{E}_{t_k}(Y|A = 1) = \frac{1}{n'} \sum_{i=1}^{n'} \text{logit}^{-1}(\hat{\gamma}_0 + \hat{\gamma}_1 + \hat{\gamma}_2 X_i)$

- $\hat{E}_{t_k}(Y|A = 0) = \frac{1}{n'} \sum_{i=1}^{n'} \text{logit}^{-1}(\hat{\gamma}_0 + \hat{\gamma}_2 X_i)$

in all n' recruited patients.

- 3 Calculate $\hat{\theta}_{t_k} = \hat{E}_{t_k}(Y|A = 1) - \hat{E}_{t_k}(Y|A = 0)$

Group Sequential Designs

- Entail analyzing the data at K different times $t_1, \dots, t_K$.

Group Sequential Designs

- Entail analyzing the data at K different times $t_1, \dots, t_K$.
- At each analysis time t_k :
 - Calculate an estimate $\hat{\theta}_{t_k}$.
 - Calculate a standardized test statistic $Z_k = Z(t_k) = \frac{\hat{\theta}_{t_k} - \theta_0}{\widehat{\text{se}}(\hat{\theta}_{t_k})}$.
 - Compare the Z_k to some critical value for that analysis.
 - Allow to stop early for efficacy and/or futility.

Group Sequential Designs

- Entail analyzing the data at K different times $t_1, \dots, t_K$.
- At each analysis time t_k :
 - Calculate an estimate $\hat{\theta}_{t_k}$.
 - Calculate a standardized test statistic $Z_k = Z(t_k) = \frac{\hat{\theta}_{t_k} - \theta_0}{\widehat{\text{se}}(\hat{\theta}_{t_k})}$.
 - Compare the Z_k to some critical value for that analysis.
 - Allow to stop early for efficacy and/or futility.

Group Sequential Design

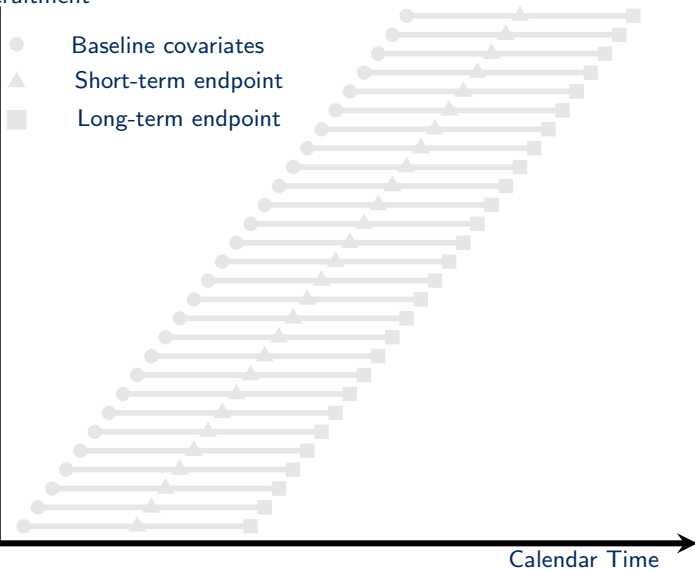
- A group sequential design can **reduce the length** of a Phase 3 trial
 - Reaching a conclusion sooner (Stopping for efficacy/futility)
 - Faster access to effective treatments
 - Faster dropping of ineffective/harmful treatment
 - **Saving resources**

Group Sequential Design

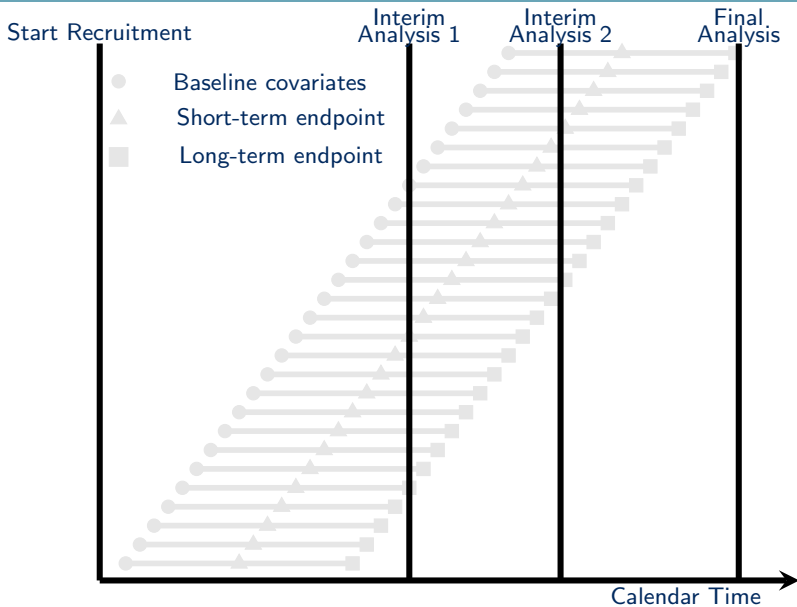
- A group sequential design can **reduce the length** of a Phase 3 trial
 - Reaching a conclusion sooner (Stopping for efficacy/futility)
 - Faster access to effective treatments
 - Faster dropping of ineffective/harmful treatment
 - **Saving resources**
 - Multiple looks at accumulating data **increase type I error**
 - Lower significant thresholds needs to be used for the interim analyses.
 - There are a range of methods for defining the critical values for interim analyses.
- (Pocock, 1977; O'Brien and Fleming, 1979; Lan and DeMets, 1983)

Independent Increments

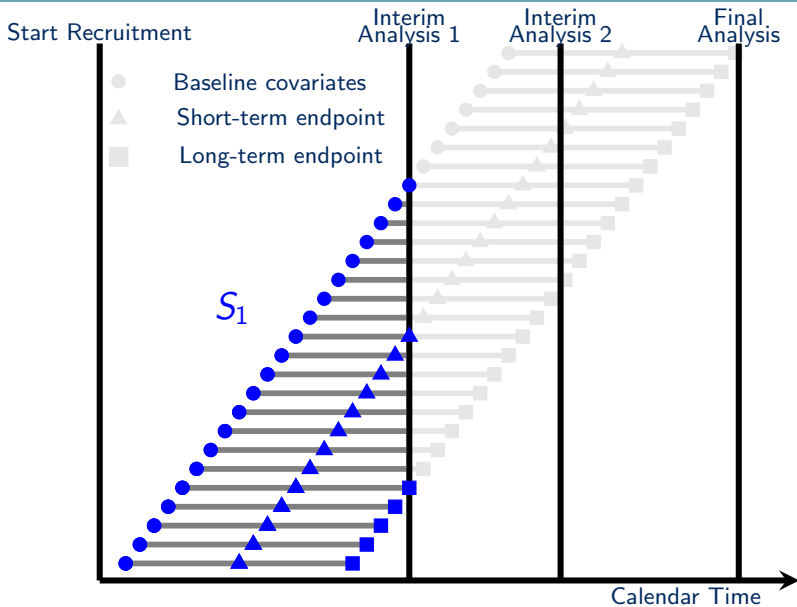
Start Recruitment



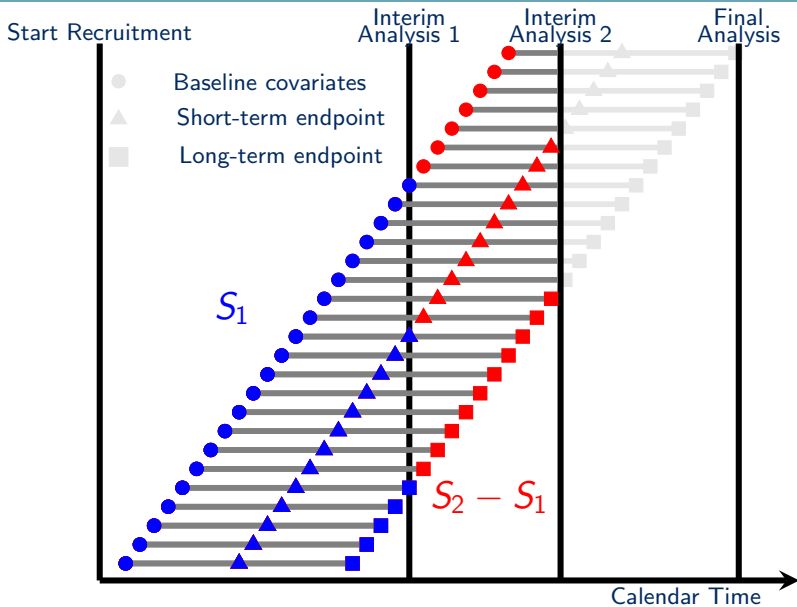
Independent Increments



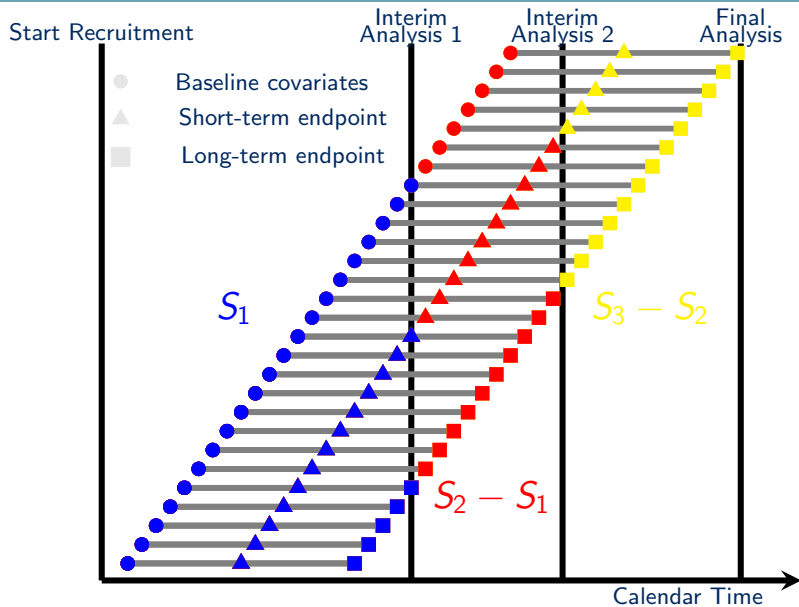
Independent Increments



Independent Increments



Independent Increments



Group Sequential Designs: Incompatibility

- Unfortunately, a sequence of RAL estimators $(\hat{\theta}_{t_1}, \dots, \hat{\theta}_{t_K})$ do not necessarily have the independent increments property.
- This was for example shown for:
 - Estimators based on generalized estimating equations (Shoben and Emerson, 2014)
 - G-computation and TMLE estimators when working models are misspecified (Rosenblum et al., 2015)
 - A long list of further counterexamples is provided by Jennison and Turnbull (1997) and Kim and Tsiatis (2020)

Group Sequential Designs: Incompatibility

- Unfortunately, a sequence of RAL estimators $(\hat{\theta}_{t_1}, \dots, \hat{\theta}_{t_K})$ do not necessarily have the independent increments property.
- This was for example shown for:
 - Estimators based on generalized estimating equations (Shoben and Emerson, 2014)
 - G-computation and TMLE estimators when working models are misspecified (Rosenblum et al., 2015)
 - A long list of further counterexamples is provided by Jennison and Turnbull (1997) and Kim and Tsiatis (2020)
- Proposal: **modifying** any RAL estimator so that it has the **independent increments property** and also has **equal or smaller variance than the original estimator**.

Proposal: Motivation

- **Goal:** Obtain at each analysis time t_k an estimator $\tilde{\theta}_{t_k}$ that
 - 1 is consistent for θ ,
 - 2 is asymptotically linear,
 - 3 is asymptotically normal,
 - 4 is asymptotically as or more precise as the original estimator $\hat{\theta}_{t_k}$, and
 - 5 has the independent increments property.

Proposal: Motivation

■ **Goal:** Obtain at each analysis time t_k an estimator $\tilde{\theta}_{t_k}$ that

- 1 is consistent for θ ,
- 2 is asymptotically linear,
- 3 is asymptotically normal,
- 4 is asymptotically as or more precise as the original estimator $\hat{\theta}_{t_k}$, and
- 5 has the independent increments property.

■ We will focus on finding the **linear combination**

$$\hat{\theta}_{t_k} - \sum_{k'=1}^{k-1} \lambda_{k'}^{(k)} (\hat{\theta}_{t_k} - \hat{\theta}_{t_{k'}})$$

with **minimal variance**.

Plan

- 1 Background, Problem Setting and Set Up
- 2 Proposal: Combining Covariate Adjustment and GSDs
- 3 Information-Adaptive Design**
- 4 Simulation Study
- 5 Discussion

Algorithm for Analysis Timing: Design Stage

- Specify the operating characteristics of the study

Algorithm for Analysis Timing: Design Stage

- Specify the operating characteristics of the study
- We **compute the maximum/total information** needed to preserve these operational characteristics

$$\left(\frac{z_{\alpha/2} + z_{\beta}}{\theta_A - \theta_0} \right)^2,$$

for a fixed design (no interim analyses), and

$$\left(\frac{z_{\alpha/2} + z_{\beta}}{\theta_A - \theta_0} \right)^2 IF$$

when data is sequentially monitored with the possibility of early stopping.

(Mehta and Tsiatis, 2001)

Algorithm for Analysis Timing: Information

- We propose to **monitor the accrued information**, $(\widehat{\text{se}}(\hat{\theta}_t))^{-2}$, through time t .

Algorithm for Analysis Timing: Information

- We propose to **monitor the accrued information**, $(\widehat{se}(\hat{\theta}_t))^{-2}$, through time t .
- We consider a trial with an interim analysis when 50% of the information is available:
 - We conduct the interim analysis at time t_1 when

$$(\widehat{se}(\hat{\theta}_{t_1}))^{-2} \geq 0.5 \cdot \left(\frac{z_{\alpha/2} + z_{\beta}}{\theta_A - \theta_0} \right)^2 IF.$$

- We conduct the final analysis at time t_2 when

$$(\widehat{se}(\hat{\theta}_{t_2}))^{-2} \geq \left(\frac{z_{\alpha/2} + z_{\beta}}{\theta_A - \theta_0} \right)^2 IF.$$

Plan

- 1 Background, Problem Setting and Set Up
- 2 Proposal: Combining Covariate Adjustment and GSDs
- 3 Information-Adaptive Design
- 4 Simulation Study**
- 5 Discussion

MISTIE III trial (Stroke)

- Functional outcome: proportion of patients who achieved a modified Rankin Scale score of 0-3 at 365 days (**binary**).
- Estimand of interest: **risk difference**.
- Total sample size of approximately 498 patients (in original trial):

MISTIE III trial (Stroke)

- Functional outcome: proportion of patients who achieved a modified Rankin Scale score of 0-3 at 365 days (**binary**).
- Estimand of interest: **risk difference**.
- Total sample size of approximately 498 patients (in original trial):
 - 1:1 randomization
 - Power of 88% to detect an average effect size of 13% at a 5% significance level
 - Success rate: 25% in standard medical care group versus 38% in MISTIE group

MISTIE III trial (Stroke)

- Functional outcome: proportion of patients who achieved a modified Rankin Scale score of 0-3 at 365 days (**binary**).
- Estimand of interest: **risk difference**.
- Total sample size of approximately 498 patients (in original trial):
 - 1:1 randomization
 - Power of 88% to detect an average effect size of 13% at a 5% significance level
 - Success rate: 25% in standard medical care group versus 38% in MISTIE group
- We will focus on information instead of sample size!

Simulation Study: $K = 1$

- **Information-adaptive design** with maximum information equal to 582
- **Maximum sample size design** with $n_{max} = 498$

		$\theta = 0.13$ (Alternative)			
		Power	ASN	AAT	AI
Information-adaptive design	Unadjusted	88.4%	571	1876	582
	Standardization	87.3%	433	1509	567
Maximum sample size design	Unadjusted	83.1%	-	1682	508
	Standardization	91.1%	-	1682	652

ASN: average sample number; AAT: average analysis time (days); AI: average information.

Conclusion under alternative:

24% reduction of sample size due to covariate adjustment

Simulation Study: $K = 1$

- **Information-adaptive design** with maximum information equal to 582
- **Maximum sample size design** with $n_{max} = 498$

		$\theta = 0$ (Null)			
		Type I	ASN	AAT	AI
Information-adaptive design	Unadjusted	5.28%	569	1871	582
	Standardization	5.28%	402	1427	568
Maximum sample size design	Unadjusted	5.14%	-	1682	509
	Standardization	5.14%	-	1682	705

ASN: average sample number; AAT: average analysis time (days); AI: average information.

Conclusion under null:

29% reduction of sample size due to covariate adjustment

Simulation Study: $K = 2$

- We perform interim analysis when 50% of the (total) information is available
 - Total information: 648

		$\theta = 0.13$ (Alternative)		
		Power	ASN	AAT
Original estimators $\hat{\theta}_{t_k}$	Unadjusted	88.3%	534	1566
	Standardization	87.1%	431	1299
Orthogonalized estimators $\tilde{\theta}_{t_k}$	Standardization	87.0%	431	1299

ASN: average sample number; AAT: average analysis time (days).

Note: We did a small sample size correction for standardization estimator.

Conclusion under alternative:

19% reduction of sample size due to covariate adjustment

Simulation Study: $K = 2$

- We perform interim analysis when 50% of the (total) information is available
 - Total information: 648

		$\theta = 0$ (Null)		
		Type I	ASN	AAT
Original estimators $\hat{\theta}_{t_k}$	Unadjusted	5.29%	628	2014
	Standardization	5.06%	449	1542
Orthogonalized estimators $\tilde{\theta}_{t_k}$	Standardization	5.05%	449	1542

AAT: average analysis time (days); ASN: average sample number.

Note: We did a small sample size correction for standardization estimator.

Conclusion under null:

29% reduction of sample size due to covariate adjustment

Plan

- 1 Background, Problem Setting and Set Up
- 2 Proposal: Combining Covariate Adjustment and GSDs
- 3 Information-Adaptive Design
- 4 Simulation Study
- 5 Discussion**

Discussion

- We performed additional simulations under violation of independent increments property:

			Type I
Original estimators $\hat{\theta}_{t_k}$	Standardization		5.37%
Orthogonalized estimators $\tilde{\theta}_{t_k}$	Standardization		5.07%

Discussion

- We performed additional simulations under violation of independent increments property:

		Type I
Original estimators $\hat{\theta}_{t_k}$	Standardization	5.37%
Orthogonalized estimators $\tilde{\theta}_{t_k}$	Standardization	5.07%

- Simulations have only shown small deviations from independent increment structure.

Discussion

- We performed additional simulations under violation of independent increments property:

			Type I
Original estimators $\hat{\theta}_{t_k}$	Standardization		5.37%
Orthogonalized estimators $\tilde{\theta}_{t_k}$	Standardization		5.07%

- Simulations have only shown small deviations from independent increment structure.
- In practice, underlying **data-generating mechanism is unknown**.

Discussion

- We performed additional simulations under violation of independent increments property:

			Type I
Original estimators $\hat{\theta}_{t_k}$	Standardization		5.37%
Orthogonalized estimators $\tilde{\theta}_{t_k}$	Standardization		5.07%

- Simulations have only shown small deviations from independent increment structure.
- In practice, underlying **data-generating mechanism is unknown**.
- **Safer** to use the proposal as it guarantees to maintain the Type I error in large samples.

Discussion

- Importantly, works for **all kind of endpoints and estimands** as long as the considered estimators are consistent and asymptotically linear (**Not necessarily covariate adjusted estimators!**).
- Our proposal will result in **faster, more efficient trials** for many disease areas, without sacrificing validity or power.
 - Can lead to faster trials even when the treatment is ineffective.

Discussion

- Importantly, works for **all kind of endpoints and estimands** as long as the considered estimators are consistent and asymptotically linear (**Not necessarily covariate adjusted estimators!**).
- Our proposal will result in **faster, more efficient trials** for many disease areas, without sacrificing validity or power.
 - Can lead to faster trials even when the treatment is ineffective.
- The approach can be extended to handle **stratified randomization** and **missing data** due to drop-out under MAR.

Thank you for your attention!

Interested? Van Lancker, K., Betz, J. F., & Rosenblum, M. (2025). Combining covariate adjustment with group sequential, information-adaptive designs to improve randomized trial efficiency. Biometrics.

E-mail: kelly.vanlancker@ugent.be

Website: kellyvanlancker.com

References I

- Benkeser, D., I. Díaz, A. Luedtke, J. Segal, D. Scharfstein, and M. Rosenblum (2020). Improving precision and power in randomized trials for covid-19 treatments using covariate adjustment, for binary, ordinal, and time-to-event outcomes. *Biometrics*.
- Jennison, C. and B. W. Turnbull (1997). Group-sequential analysis incorporating covariate information. *Journal of the American Statistical Association* 92(440), 1330–1341.
- Jiang, F., L. Tian, H. Fu, T. Hasegawa, and L. J. Wei (2018). Robust alternatives to ANCOVA for estimating the treatment effect via a randomized comparative study. *Journal of the American Statistical Association* 0, 1–37.
- Kim, K. and A. A. Tsiatis (2020). Independent increments in group sequential tests: a review. *SORT-Statistics and Operations Research Transactions*, 223–264.

References II

- Koch, G. G., C. M. Tangen, J.-W. Jung, and I. A. Amara (1998). Issues for covariance analysis of dichotomous and ordered categorical data from randomized clinical trials and non-parametric strategies for addressing them. *Stat. Med.* 17(15-16), 1863–1892.
- Lan, G. K. and D. L. DeMets (1983). Discrete sequential boundaries for clinical trials. *Biometrika* 70(3), 659–663.
- Mehta, C. R. and A. A. Tsiatis (2001). Flexible sample size considerations using information-based interim monitoring. *Drug information journal: DIJ/Drug Information Association* 35(4), 1095–1112.
- Moore, K. and M. J. van der Laan (2009a). Covariate adjustment in randomized trials with binary outcomes: Targeted maximum likelihood estimation. *Stat. Med.* 28(1), 39–64.

References III

- Moore, K. L. and M. J. van der Laan (2009b). Increasing power in randomized trials with right censored outcomes through covariate adjustment. *Journal of Biopharmaceutical Statistics* 19(6), 1099–1131. PMID: 20183467.
- O'Brien, P. C. and T. R. Fleming (1979). A multiple testing procedure for clinical trials. *Biometrics*, 549–556.
- Pocock, S. J. (1977). Group sequential methods in the design and analysis of clinical trials. *Biometrika* 64(2), 191–199.
- Rosenblum, M., T. Qian, Y. Du, , and H. Qiu (2015). Adaptive enrichment designs for randomized trials with delayed endpoints, using locally efficient estimators to improve precision. *Johns Hopkins University, Dept. of Biostatistics Working Papers*. <https://biostats.bepress.com/jhubiostat/paper275>.

References IV

- Rubin, D. and M. van der Laan (2008). Covariate adjustment for the intention-to-treat parameter with empirical efficiency maximization. *U.C. Berkeley Division of Biostatistics Working Paper Series. Working Paper 229*, <https://biostats.bepress.com/ucbbiostat/paper229>.
- Shoben, A. B. and S. S. Emerson (2014). Violations of the independent increment assumption when using generalized estimating equation in longitudinal group sequential trials. *Statistics in medicine* 33(29), 5041–5056.
- Tsiatis, A. A., M. Davidian, M. Zhang, and X. Lu (2008). Covariate adjustment for two-sample treatment comparisons in randomized clinical trials: a principled yet flexible approach. *Statistics in medicine* 27(23), 4658–4677.

References V

- Yang, L. and A. Tsiatis (2001). Efficiency study of estimators for a treatment effect in a pretest-posttest trial. *The American Statistician* 55(4), 314–321.
- Zhang, D. (2009). Lecture notes for statistical principles of clinical trials (modified from dr. a. tsiatis' lecture notes).
- Zhang, M. (2015, Jan). Robust methods to improve efficiency and reduce bias in estimating survival curves in randomized clinical trials. *Lifetime Data Analysis* 21(1), 119–137.